ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S [ANSWER KEY]

Unit 5.01: Variation of Parameters & Greens Kernel

Q1. Find a particular solution of y''-y=2 using variation of parameters.

[ ] A: 2
[x] B: -2
[ ] C: -1
[ ] D: -tfrac12

Solution: y_1=e^{x} , y_2=e^{-x} , W=-2 . u_1'=-dfrac{y_22}{W}=e^{-x}Rightarrow u_1=-e^{-x} ; u_2'=dfrac{y_12}{W}=-e^{x}Rightarrow u_2=-e^{x} . So y_p=u_1y_1+u_2y_2=-e^{-x}e^{x}-e^{x}e^{-x}=-2 .

Q2. Find a particular solution of y''-4y'+4y=6 using variation of parameters.

[ ] A: -tfrac32
[ ] B: 3
[x] C: tfrac32
[ ] D: 6

Solution: y_1=e^{2x} , y_2=xe^{2x} repeated root, W=e^{4x} . u_1'=-dfrac{y_26}{W}=-6xe^{-2x} , u_2'=dfrac{y_16}{W}=6e^{-2x} ; integrating and substituting back gives y_p=tfrac32 constant â€" matching what undetermined coefficients would give directly, A=6/4 .

Q3. Find a particular solution of y''+4y=8 using variation of parameters.

[x] A: 2
[ ] B: -2
[ ] C: 4
[ ] D: 8

Solution: y_1=cos2x , y_2=sin2x , W=2 . u_1'=-dfrac{y_28}{W}=-4sin2xRightarrow u_1=2cos2x ; u_2'=dfrac{y_18}{W}=4cos2xRightarrow u_2=2sin2x . So y_p=u_1y_1+u_2y_2=2cos^22x+2sin^22x=2 .

Q4. Find a particular solution of y''+y=cot x for 0<x<pi using variation of parameters.

[ ] A: sin xln|csc x+cot x|
[ ] B: -cos xln|csc x+cot x|
[ ] C: -sin xln|sec x+tan x|
[x] D: -sin xln|csc x+cot x|

Solution: y_1=cos x , y_2=sin x , W=1 . u_1'=-sin xcot x=-cos xRightarrow u_1=-sin x ; u_2'=cos xcot x=csc x-sin xRightarrow u_2=cos x-ln|csc x+cot x| . So y_p=u_1y_1+u_2y_2=-sin xcos x+bigcos x-ln|csc x+cot x|bigsin x=-sin xln|csc x+cot x| the sin xcos x terms cancel.

Q5. Find a particular solution of y''-4y=e^{2x} using variation of parameters.

[ ] A: Bigtfrac{x}{4}+tfrac{1}{16}Bige^{2x}
[x] B: Bigtfrac{x}{4}-tfrac{1}{16}Bige^{2x}
[ ] C: Bigtfrac{x}{2}-tfrac{1}{16}Bige^{2x}
[ ] D: -Bigtfrac{x}{4}-tfrac{1}{16}Bige^{2x

Solution: y_1=e^{2x} , y_2=e^{-2x} , W=-4 . Here r=2 matches a root â€" no modification is needed for variation of parameters, unlike undetermined coefficients. u_1'=dfrac14Rightarrow u_1=dfrac{x}{4} ; u_2'=-dfrac{e^{4x}}{4}Rightarrow u_2=-dfrac{e^{4x}}{16} . So y_p=u_1y_1+u_2y_2=Bigdfrac{x}{4}-dfrac{1}{16}Bige^{2x} .

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Q6. Find a particular solution of y''+4y=csc2x for 0<x<pi/2 using variation of parameters.

[x] A: -tfrac{x}{2}cos2x+tfrac14sin2xln|sin
[ ] B: tfrac{x}{2}cos2x+tfrac14sin2xln|sin2
[ ] C: -tfrac{x}{2}cos2x-tfrac14sin2xln|sin
[ ] D: -tfrac{x}{4}cos2x+tfrac14sin2xln|sin

Solution: y_1=cos2x , y_2=sin2x , W=2 . u_1'=-dfrac12Rightarrow u_1=-dfrac{x}{2} ; u_2'=dfrac12cot2xRightarrow u_2=dfrac14ln|sin2x| . So y_p=u_1y_1+u_2y_2=-dfrac{x}{2}cos2x+dfrac14sin2xln|sin2x| .

Q7. Find a particular solution of y''-6y'+9y=dfrac{e^{3x}}{x^2} for x>0 using variation of parameters.

[ ] A: e^{3x}ln x+1
[ ] B: -e^{3x}ln x-1
[x] C: -e^{3x}ln x+1
[ ] D: -e^{3x}ln x

Solution: y_1=e^{3x} , y_2=xe^{3x} repeated root, W=e^{6x} . u_1'=-dfrac1xRightarrow u_1=-ln x ; u_2'=dfrac{1}{x^2}Rightarrow u_2=-dfrac1x . So y_p=u_1y_1+u_2y_2=-e^{3x}ln x-e^{3x}=-e^{3x}ln x+1 .

Q8. Find a particular solution of y''+4y'+3y=e^{-x} using variation of parameters.

[ ] A: Bigtfrac{x}{2}+tfrac14Bige^{-x}
[ ] B: x-tfrac14e^{-x}
[ ] C: -Bigtfrac{x}{2}-tfrac14Bige^{-x}
[x] D: Bigtfrac{x}{2}-tfrac14Bige^{-x}

Solution: y_1=e^{-x} , y_2=e^{-3x} roots -1,-3 , W=-2e^{-4x} . u_1'=dfrac12Rightarrow u_1=dfrac{x}{2} ; u_2'=-dfrac{e^{2x}}{2}Rightarrow u_2=-dfrac{e^{2x}}{4} . So y_p=u_1y_1+u_2y_2=Bigdfrac{x}{2}-dfrac14Bige^{-x} .